

Rohan Karamel

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EDUCATION

Rutgers University

Bachelor of Science in Computer Science and Mathematics

Expected December 2025

New Brunswick, NJ

- **Cumulative GPA:** 3.7/4.0
- **Leadership Roles:** Rutgers AI & DS Collaboratory Ambassador, IEEE ML/AI Division Lead, Rutgers Poker Club Treasurer

WORK EXPERIENCE

Rutgers AI and Data Science Collaboratory

Research Fellow

May 2025 – Present

New Brunswick, NJ

- Developed predictive models of neuron deformation under injury conditions to better understand **traumatic brain injury**.
- Automated large-scale **Abaqus** simulations in **Python** to analyze white-matter brain data across **thousands** of 3D geometries.
- Trained **recurrent neural networks** to model behavior from simulation data, enabling characterization of cellular material properties.
- Co-authoring a paper titled 'A Machine Learning-based Modeling of Fractional Viscoelasticity in Brain White Matter' (in preparation).

General Electric Aerospace

Machine Learning Engineer Intern

May 2024 – August 2024

Austin, TX

- Created datasets with classifications of **3,000,000+** historical flight scheduling problems for major airlines using **Java** and **Azure**.
- Cut average solve time by over **50%** and reduced solver costs by **\$4,000/month** using an **XGBoost** classifier to recommend the most historically effective resolution strategy with **90% accuracy**.
- Delivered **thousands** of **real-time** operational strategies to **air traffic controllers** for disruption recovery problems **every day**.
- Laid the foundation for **ongoing research** initiatives that continued beyond my internship, shaping the next phase of the project.

PROJECTS

GPU Mandelbrot Renderer

Swift, C, Metal, GPU Programming

github.com/rakdcolon/MandelbrotGPU

- Built interactive fractal explorer with **GPU shaders** on **Apple Silicon Metal API**, supporting real-time zoom/pan at **16K resolution**.
- Optimized GPU memory usage and **parallel execution**, achieving high frame rates while handling fractals with precision rendering.

JavaLM

Java, Large Language Models, Transformer Architecture, NLP, Deep Learning

github.com/rakdcolon/JLLM

- Designed a **125M**-parameter transformer (GPT-2) in pure **Java** with performance tuning (**39.2 perplexity**, **120 tokens/s**).
- Implemented the transformer architecture: **multi-head self-attention**, **positional embeddings**, and **feed-forward networks**.

MyOperatingSystem

C, C++, Linux, Memory Management, Operating Systems

github.com/rakdcolon/Operating-Systems

- Developed core **operating system** components, including a custom **Malloc Library**, **Thread Library**, **Shell** in **C/C++**.
- Introduced a **page table system** and **FUSE-based file system** for virtual memory and file management.

RELEVANT COURSEWORK

Graduate: Machine Learning, Intro to Artificial Intelligence, Linear Algebra Applications, High-Dimensional Expanders and Graphs

Undergraduate: Deep Learning, Computational Robotics, Data Science, Operating Systems, Formal Language & Automata, Algorithms, Numerical Analysis, Systems Programming, Computer Architecture, Data Structures, Statistics, Graph Theory, Combinatorics, Probability, Real Analysis, Abstract Algebra, Linear Algebra, Multivariable Calculus, Differential Equations

SKILLS

Languages: Python, C++, Java, C, MATLAB, Bash, XML, JSON

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Matplotlib, Pandas, Seaborn, Metal, OpenCV, Abaqus, YFinance

Tools & Cloud: Azure, Git, Linux, Unix, PowerBI, Jupyter, GitHub, Azure DevOps, Docker

Concepts: Deep Learning, Transformers, Time-Series Modeling, Natural Language Processing, Neural Networks, GPU Programming, Object-Oriented Programming, Data Structures & Algorithms, Software Development